

A 40 year old man with hypertrophic obstructive cardiomyopathy (HOCM) presents for elective laparoscopic cholecystectomy.

(a) Describe the principles of intraoperative haemodynamic management for this patient. (40%)

(b) How would you manage hypotension post induction of general anaesthesia in this patient? (60%)

(Modified from ANZCA 2010A)

Examiner Report

67.5% of candidates passed this question

The following were key components of an answer required to pass this question:

Part A.

- Intraoperative haemodynamic management is based upon the severity of the condition and the consequences of the procedure.
- To pass, a candidate was required to demonstrate an understanding of the pathophysiology of this disease, particularly the consequences and management of the dynamic obstruction, and the need to minimise obstruction by having a full ventricle (high preload and high systemic vascular resistance), slow heart rate, low contractility.

Full marks required a discussion of:

- severity of disease including past and current treatment, recent investigations including echocardiogram.
- effects of systolic anterior motion of anterior leaflet of mitral valve
- existing therapy such as beta blockers and AICD,
- left ventricular hypertrophy and diastolic dysfunction
- risk of arrhythmias (including of the common ones i.e. VT/VF as a cause of sudden death; atrial fibrillation)
- physiological consequences of the procedure (pressure from pneumoperitoneum, position of patient, carbon dioxide absorption).

Part B

A pass required the following:

- A differential diagnosis, including cardiac and non-cardiac causes. Typically candidates limited their answer to exacerbation of left ventricular outflow tract (LVOT) obstruction, myocardial ischaemia, and to anaphylaxis.
- Appropriate therapy: alpha adrenergic agonists, volume loading, beta adrenergic blockade, reduction in sympathetic tone (including adequate analgesia with opioid administration) and avoiding beta adrenergic sympathetic stimulation in the pharmacological management. Most candidates based their treatment on the assumption that the hypotension was induced by LVOT obstruction after drug induced loss of afterload (through vasodilatation) and loss of preload (through vasodilatation).
- If immediately life-threatening, an ABC approach was needed, with discontinuation of the anaesthetic and administration of 100% oxygen.

Full marks required:

- The simultaneous diagnosis and treatment based upon severity of the clinical problem, including determining if the problem is life threatening, in which case the surgeon should be advised of the problem, a crisis should be declared and help should be called.
- Discussion of rhythm disturbances and the indications for cardioversion.
- Detailed outline of myocardial blood supply/ demand relationship in this context.
- The potential for the hypotension to be related to the procedure if it had begun, and the need to release any pneumoperitoneum and to change the position from reverse-Trendelenburg to Trendelenburg.
- How the diagnosis might be made using readily available clinical signs, monitoring decisions and potential to use echocardiography.

Comments about how this question was answered:

- Many candidates found this question difficult. Many failed to recognize that there were two potential problems to address. In the first part it was important to discuss the haemodynamic effects of both HOCM and laparoscopic cholecystectomy.
- Most candidates passed part A, but then many failed the question as a whole by providing insufficient information (in many cases almost none) in Part B.
- In Part B, it was not sufficient to merely re-state the information from Part A. In particular, many candidates failed to create a differential diagnosis in Part B. This made it difficult to provide a framework for a structured answer.

Sample Answer

(a)

Principles of intra-operative haemodynamic management would be dependent on:

Severity of HOCM

- AICD in-situ (suggesting risk of malignant arrhythmias)
- Past and current treatment, including B-blockers
- ECG: LVH, arrhythmias
- 2DE findings: Assess biventricular contractility, degree of LVOT obstruction, presence of Systolic Anterior Motion (SAM), diastolic dysfunction, volume status

Effect of surgery on HOCM

- Pneumoperitoneum (Increased IAP reducing venous return/ preload/ vagal stimulation)
- CO₂ absorption (may cause tachycardia/arrhythmias)
- Reverse Trendelenburg position (decrease preload)

Principles of intra-operative haemodynamic management for this patient would include:

Avoid worsening dynamic LVOT obstruction

- Use of TEE assessment of biventricular contractility
- Keeping the LV well-filled by maintaining
 - High-normal preload
 - Adequate IV fluid loading before Reverse Trendelenburg position/pneumoperitoneum
 - Goal directed fluid therapy using cardiac output monitor (Flotrac)
 - High-normal afterload/ systemic vascular resistance
 - Vasopressors (Phenylephrine, Vasopressin) to maintain BP during induction
 - Avoid histamine releasing drugs (Rocuronium over Atracurium, Oxycodone over Morphine)
- Maintain low-normal contractility
- Maintain low-normal heart rate

Optimise coronary perfusion and myocardial O₂ supply

- Check and treat anaemia
- Titrate FiO₂/PEEP to keep SpO₂>95%
- Maintain diastolic BP (high-normal afterload), diastolic time (slow HR) and filling to maintain coronary perfusion pressure

Avoid surges in myocardial O₂ demand

- Continue beta-blockers perioperatively if patient is already on them
- Avoid sympathetic surges
 - Blunt laryngoscopy response with esmolol/lignocaine/alfentanil
 - Adequate depth of anaesthesia and analgesia
 - Titrate ventilation to aim ET CO₂ 35-40 mmHg (avoid hypercarbia)
 - Smooth extubation (on remifentanil)
- Treat tachyarrhythmias aggressively

Maintain sinus rhythm, with standby measures for malignant arrhythmias

- Check and correct electrolyte abnormalities/acidosis
- Ensure normothermia (temp probe monitoring, active warming devices)

- Peri-op AICD management (if present), defibrillation pads, resus drugs drawn, crash cart on standby

(b)

I would manage post-induction hypotension based on severity of clinical problem. If hypotension is profound and potentially life-threatening, I would declare crisis situation, inform surgeon, and call for help (including defib pads and crash cart).

Possible causes include:

Patient-related	Worsening dynamic LVOT obstruction Cardiac events (ischaemia, arrhythmia)
Anaesthesia-related	Drop in SVR from induction agent Anaphylaxis Tension pneumothorax
Surgical-related	Pneumoperitoneum (if initiated) Bleeding (if surgery has started)

I would simultaneously resuscitate patient according to ABC principles, and assess for other possible causes of hypotension. In this patient with history of HOCM, I would be most concerned about worsening dynamic LVOT obstruction.

Airway	Check and ensure correct ETT placement and patency
Breathing	Give 100% FiO ₂ . Auscultate to <u>exclude tension pneumothorax</u> (absent breath sound, tracheal deviation, hyperresonance to percussion) , <u>anaphylaxis</u> (wheeze), <u>pulmonary oedema</u> (creps)
Circulation	Feel for pulse. Check other haemodynamic parameters (HR, SpO ₂ , ETCO ₂) for abnormalities. Check ECG for <u>ischaemic changes</u> and <u>arrhythmias</u> Vasopressors (<u>Phenylephrine</u>) to improve BP/afterload Run <u>IV crystalloids</u> fast via 2x large bore IV cannula <u>Stop/ cut down</u> on anaesthetic agents <u>Release pneumoperitoneum</u> and <u>level</u> patient (if surgery has started)

Subsequent treatment should be directed at the cause of hypotension. Specific treatment for worsening dynamic LVOT obstruction include:

- Increase afterload with direct α -agonist (phenylephrine)
- Increase preload by fluid resuscitation with IV crystalloids
- B-blocker to slow down HR and reduce contractility
- Treat arrhythmias aggressively (correct electrolytes, cardioversion)
- Treat causes of sympathetic stimulation (pain, hypercarbia) by optimising depth of anaesthesia/ analgesia and minute ventilation
- TEE to evaluate biventricular contractility, degree of LVOT obstruction, presence of SAM, diastolic dysfunction, volume status

I would discuss with surgeon regarding continuation vs stopping surgery if patient remains unstable. Post-operative disposition to HD/ICU for monitoring. Cardiology referral for work-up and optimisation when in HD/ICU